THE NOVA RESEARCH CURRICULUM

Answer key · Grade 8 · Term 3

For teachers only. Do not share with students.

Teacher answer key

English

Question bank

#	Outcome	Question	Answer
1	EN-8.4	"Shouldn't every child have clean water?" is a ...	simile / **rhetorical question** / metaphor / quotation
2	EN-8.4	"We will plan, we will act, we will succeed" uses ...	alliteration / **the rule of three** / irony / a pun
3	EN-8.4	"Show, don't tell" means ...	use more pictures / **let actions and details reveal feelings** / write less / tell the ending first
4	EN-8.4	A short story should stay in one point of view unless there is a reason to change. True or false.	**True**
5	EN-8.4	Rewrite "He was scared" using show, don't tell.	Model: His hands shook and he could hear his heart thumping as the door creaked open.
6	EN-8.5	A good discussion leader ...	speaks the most / **invites everyone to speak** / ends early / chooses one side
7	EN-8.5	Facts in a persuasive speech should come from reliable sources. True or false.	**True**

Urdu

Question bank

#	Outcome	Question	Answer
1	UR-8.5	تلخیص میں کیا شامل نہیں کیا جاتا؟	بنیادی نکات / موزوں عنوان / **مثالیں اور تکرار** / اپنے الفاظ
2	UR-8.5	تلخیص کا ایک موزوں عنوان ہوتا ہے۔ True or false.	**True**
3	UR-8.5	تحقیقی نوٹ کے آخر میں کیا لکھا جاتا ہے؟	شعر / **حوالہ جات** / مکالمہ / محاورہ
4	UR-8.6	پُراثر تقریر کے تین حصے لکھیں۔	Model: آغاز، دلائل، پرجوش اختتام۔

#	Outcome	Question	Answer
5	UR-8.6	True or false. شعر خوانی میں درست تلفظ ضروری ہے۔	**True**

Mathematics

Question bank

#	Outcome	Question	Answer
1	MA-8.4	Volume of a pyramid with a 6 cm × 6 cm base and height 10 cm?	360 cm³ / **120 cm³** / 60 cm³ / 180 cm³
2	MA-8.4	Volume of a cone with r = 3 cm, h = 7 cm (π = 22/7)?	198 cm³ / **66 cm³** / 22 cm³ / 132 cm³
3	MA-8.4	Surface area of a sphere with r = 7 cm (π = 22/7)?	154 cm² / 308 cm² / **616 cm²** / 1,437 cm²
4	MA-8.4	A cone holds one third of a cylinder with the same base and height. True or false.	**True**
5	MA-8.6	Range of 4, 9, 2, 15, 7?	11 / **13** / 15 / 7
6	MA-8.6	Data 2, 4, 4, 4, 5, 5, 7, 9 has mean 5. Its standard deviation?	1 / **2** / 4 / 5
7	MA-8.6	A coin landed heads 18 times in 40 tosses. Experimental P(heads)?	0.5 / **0.45** / 0.18 / 0.4
8	MA-8.6	With more trials, experimental probability usually gets closer to theoretical probability. True or false.	**True**
9	MA-8.6	A frequency polygon joins the ... of each class.	lower limits / **midpoints** / upper limits / totals

Physics

Question bank

#	Outcome	Question	Answer
1	PH-8.3	A 200 g block has volume 50 cm³. Its density?	250 g/cm³ / **4 g/cm³** / 0.25 g/cm³ / 150 g/cm³
2	PH-8.3	An object with density 0.8 g/cm³ in water will ...	sink / **float** / dissolve / explode
3	PH-8.3	A 600 N force acts on 2 m². Pressure?	1,200 Pa / **300 Pa** / 602 Pa / 3 Pa
4	PH-8.3	Dams are thicker at the bottom because liquid pressure increases with depth. True or false.	**True**
5	PH-8.3	Car brakes use ...	magnets / **hydraulics** / sound / solar power

#	Outcome	Question	Answer
6	PH-8.3	If the forces on a moving car are balanced, it …	stops at once / **keeps moving at the same speed** / speeds up / turns
7	PH-8.4	Our galaxy is called the …	Andromeda / **Milky Way** / Solar System / Orion
8	PH-8.4	A star like the Sun will end as a …	black hole / **white dwarf** / comet / planet
9	PH-8.4	A light-year is a unit of …	time / **distance** / brightness / mass

Chemistry

Question bank

#	Outcome	Question	Answer
1	CH-8.4	A pH of 3 is …	neutral / alkaline / **acidic** / impossible
2	CH-8.4	Litmus turns … in an alkali.	red / **blue** / green / yellow
3	CH-8.4	Pure water has pH …	0 / 1 / **7** / 14
4	CH-8.4	Acid + base → …	gas only / **salt + water** / metal + oxygen / acid
5	CH-8.4	Antacid tablets neutralise extra stomach acid. True or false.	**True**
6	CH-8.4	Which is an alkali?	vinegar / lemon juice / **soap solution** / cola
7	CH-8.4	Name the salt made from HCl and NaOH.	Model: sodium chloride (NaCl), common salt.
8	CH-8.4	Farmers add lime to soil that is too …	alkaline / **acidic** / wet / dry

Biology

Question bank

#	Outcome	Question	Answer
1	BI-8.3	Which process removes carbon dioxide from the air?	respiration / burning / **photosynthesis** / decay
2	BI-8.3	Organisms that break down dead material are …	producers / **decomposers** / consumers / predators
3	BI-8.3	Which is a greenhouse gas?	oxygen / nitrogen / **methane** / argon
4	BI-8.3	Cutting down forests increases carbon dioxide in the air. True or false.	**True**
5	BI-8.3	Insulin for diabetes can now be made by …	plants / **engineered bacteria** / rocks / yeast only

#	Outcome	Question	Answer
6	BI-8.3	Yoghurt is made using ...	yeast / **bacteria** / viruses / fungi only
7	BI-8.3	Give two effects of global warming on Pakistan.	Model: melting glaciers and more floods (also: heatwaves, water shortages).
8	BI-8.3	All of the energy at one level passes to the next level. True or false.	**False**

History, Geography and Pakistan Studies

Question bank

#	Outcome	Question	Answer
1	HI-8.2	The Lahore Resolution was passed on ...	14 August 1947 / **23 March 1940** / 30 December 1906 / 29 December 1930
2	HI-8.2	Who presented the Lahore Resolution?	Allama Iqbal / **A. K. Fazlul Huq** / Sir Syed / Liaquat Ali Khan
3	HI-8.2	The plan that set out the partition was the ...	Cabinet Mission Plan / **3 June Plan** / Fourteen Points / Lucknow Pact
4	HI-8.2	The Quaid-e-Azam was Pakistan's first Governor-General. True or false.	**True**
5	GE-8.2	Smog is a serious winter problem in ...	Gilgit / **Lahore** / Gwadar / Skardu
6	GE-8.2	Give two ways to reduce plastic pollution.	Model: use cloth bags and recycle plastic bottles (also: refuse straws, organise clean-ups).
7	PS-8.2	Pakistan's national language is ...	English / Punjabi / **Urdu** / Sindhi
8	PS-8.2	CPEC is a project between Pakistan and ...	Iran / Turkey / **China** / Saudi Arabia
9	PS-8.2	Pakistan is a founding member of the OIC. True or false.	**True**

Islamiyat and Ethics

Question bank

#	Outcome	Question	Answer
1	IS-8.3	The word "algebra" comes from the work of ...	Ibn Sina / **Al-Khwarizmi** / Al-Biruni / Ibn al-Haytham
2	IS-8.3	Ibn Sina is famous for his book on ...	astronomy / **medicine** / poetry / farming
3	IS-8.3	Al-Biruni measured the Earth's radius near ...	Baghdad / Cairo / **Jhelum (Nandana)** / Makkah

#	Outcome	Question	Answer
4	IS-8.3	Ibn al-Haytham is known for his work on ...	**light and optics** / music / trade / law
5	IS-8.4	"There is no compulsion in religion" is from Surah ...	Al-Hujurat / **Al-Baqarah** / Luqman / Al-Mulk
6	IS-8.4	Al-A'raf 7:31 tells us to eat and drink but not to ...	share / pray / **waste** / cook
7	IS-8.4	Sharing news online without checking is fine in Islam. True or false.	**False**
8	IS-8.4	Give one way a Muslim student can protect the environment.	Model: save water during wudu and avoid wasting food (also: plant trees, recycle).

Computer Science and skills

Question bank

#	Outcome	Question	Answer
1	CS-8.3	Before using an AI answer in your work, you should ...	copy it / **check it against reliable sources** / share it / delete it
2	CS-8.3	It is safe to type your home address into an AI chatbot. True or false.	**False**
3	CS-8.3	An AI-use note says ...	nothing / **what you asked, what you used and what you changed** / your password / your marks
4	CS-8.3	Improve the prompt "tell me about rivers".	Model: "List the five longest rivers in Pakistan with their lengths, in a table, for a Grade 8 student."
5	CS-8.4	A deepfake is ...	a real news photo / **fake video or audio made to look real** / a computer virus / a type of password
6	CS-8.4	Give one sign that a video might be a deepfake.	Model: the lips do not match the words (also: strange blinking, odd shadows, unknown source).
7	CS-8.4	AI can repeat bias from its training data. True or false.	**True**
8	FL-8.1	A pitch should clearly explain the product, the customer and the price. True or false.	**True**